Persona-Aware Response Ranking Prompt

You are evaluating two different responses from the same AI model to determine which better embodies a specific persona in a dialogue scenario.

Persona Information:
- Name: {persona_name}
- Role: {persona_role}
- Expected Emotion: {expected_emotion}
- Expected Response Style: {expected_response_style}

Context:

{context}

Response A (Steered):

{steered_response}

Response B (Non-steered):

{non_steered_response}

Evaluation Criteria: Decide which response is superior for each dimension. Choose either Response A (steered) or Response B (non-steered).

1. **Trait Adherence**: Which response better matches the expected personality traits and emotional state?
   - Consider how well each response reflects the persona's characteristics
   - Evaluate alignment with the expected emotion
2. **Role Consistency**: Which response better maintains the character's role and identity?
   - Consider consistency with the persona's background and position
   - Evaluate how well the role is embodied
3. **Response Appropriateness**: Which response better matches the expected response style and context?
   - Consider adherence to the specified communication style
   - Evaluate appropriateness of tone, approach, and context
4. **Insightfulness**: Which response demonstrates more depth, thoughtfulness, and analytical reasoning?
   - Consider the level of insight and understanding shown
   - Evaluate the quality of reasoning and reflection
5. **Overall Quality**: Considering all factors, which response is better overall?
   - Make a holistic judgment considering all criteria
   - Determine which response would be more effective in this dialog context

Response Format: Return a JSON object using "A" for Response A and "B" for Response B.

```
{
  "trait_adherence": "A" or "B",
  "role_consistency": "A" or "B",
  "response_appropriateness": "A" or "B",
  "insightfulness": "A" or "B",
  "overall": "A" or "B",
  "reasoning": "Detailed explanation comparing both responses, citing specific aspects for
      each criterion."
}
```

Figure 14: Persona-aware response ranking prompt used to compare steered and non-steered outputs along trait alignment, role fidelity, contextual appropriateness, insightfulness, and overall quality.

Algorithm: PERSONA-FLOW (Predict–then–Steer)

Inputs: base LLM $\mathcal{M}$; pole vectors $\{v_{d^+}, v_{d^-}\}_{d \in \{\mathrm{O,C,E,A,N}\}}$ from PERSONA-BASE; target layer $l^\star$
Hyperparams: clip bound $\alpha_{\max}$ (default 2.0); gate τ (default 0.5); normalization scheme s

For each turn t with context C_t:

1. $\hat{\alpha}_d \leftarrow$ PREDICTCOEFFS(C_t, *persona*) for $d \in \{\mathrm{O, C, E, A, N}\}$ // signed, context-conditioned
2. $\alpha_d \leftarrow \mathrm{clip}(\hat{\alpha}_d,\ -\alpha_{\max},\ \alpha_{\max})$ // clip to $[-2, 2]$
3. **if** $|\alpha_d| < \tau$ **then** $\alpha_d \leftarrow 0$ // sparsity gate
4. Choose pole $p(d) \leftarrow (+)$ if $\alpha_d \geq 0$ else $(-)$; set $\tilde{v}_{d^{p(d)}} \leftarrow \mathrm{Norm}(v_{d^{p(d)}}; s)$
5. $v_{\text{comp}} \leftarrow \sum_d |\alpha_d| \cdot \tilde{v}_{d^{p(d)}}$ // composite steering vector
6. **Layer injection:** $h_{l^\star} \leftarrow h_{l^\star} + v_{\text{comp}}$ during decoding for turn t // residual add
7. Generate the response with $\mathcal{M}$ using the modified activations.

Figure 15: Predict–then–steer loop with coefficient clipping, magnitude gating, pole selection, normalization, and single-layer residual injection. Defaults align with §2.4: coefficients in $[-2, 2]$, gate τ=0.5, unit-norm vectors, injection at $l^\star$ from PERSONA-BASE.